Lab 4

Who Is It, and What Does It Resist?

Day 2 · 08:50–10:30 · 100 min
Pairs · swap driver and navigator at Part D

LO4: detect AMR genes and plasmids, type the isolate, and interpret the results alongside phenotypic AST

Open your handout

Lab 4 handout

`Lab4_species_typing_amr.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

The scenario

- A patient develops a **deep SSI 9 days** after an open fracture repair
- The lab reports *S. aureus*, **methicillin-resistant**
- The isolate was sequenced: you have its **assembly** (FASTA)

KEY MESSAGE

What can the genome add?

Teaching scenario. The genome is real public data from the Galaxy Training Network.

Plan for this lab

Part	What	Time
A	Get the genome	5 min
B	Who is it? Species and strain type	20 min
C	What does it resist? AMR genes	35 min
D	Is resistance mobile? Plasmids	15 min
E	Does it match the lab’s AST?	20 min

PART A

5 min

Get the genome

1. New history: `Lab4 – AMR`
2. **Upload** → **Paste/Fetch data** → paste the URL **from your handout**
3. **Type:** `fasta` · **Name:** `MRSA_contigs.fasta`

> **OPTIONAL**
>
> Your own Lab 2 assembly finished? Run every step on it too.

MLST: the strain type

Tool **MLST** → `MRSA_contigs.fasta` → scheme **automatic** → **Run**

```
MRSA_contigs.fasta   saureus   ST?   arcC(?) aroE(?) …
```

CHECK YOURSELF

1. Which **scheme** did MLST choose? What does it tell you?
2. What is the **ST**? Is it a well-known MRSA clone?

PART B2 · OPTIONAL

Species check with Kraken2

- Tool **Kraken2** on `MRSA_contigs.fasta`, with a **report**
- What is the top species?

> **KEY MESSAGE**
>
> **If the organism is wrong or the culture is mixed, every result after this is wrong.**

AMRFinderPlus (NCBI)

- **Input type:** nucleotide → `MRSA_contigs.fasta`
- **Organism:** `Staphylococcus_aureus` (finds point mutations too)
- Everything else: defaults → **Run** → open the **report**

KEY COLUMNS

Element symbol (gene) · **Class** (drug class) · **% Identity** · **% Coverage**

PART C2

staramr

- ResFinder + PointFinder + PlasmidFinder + MLST in one tool
- Input `MRSA_contigs.fasta` → defaults → **Run**
- Open **summary.tsv** and **detailed_summary.tsv**

PART C3

Record your findings

Gene / mutation	Drug class	AMRFinderPlus?	staramr?	On a plasmid?

Fill in the full table in your handout. The last column comes in Part D.

PART C

Check yourself: AMR genes

CHECK YOURSELF

1. Which gene explains the **methicillin resistance**? *(Glossary cheat sheet!)*
2. Did both tools find the same genes? If not, why might they differ?
3. What is staramr's **predicted phenotype**?

Swap driver and navigator

The navigator takes the keyboard; the driver reads the handout

Is resistance mobile? Plasmids

1. In `detailed_summary.tsv`, find the **plasmid** rows (`rep…`) and note their **contig**
2. Mark resistance genes on the **same contig** as a plasmid

ALTERNATIVE

ABRicate with the **plasmidfinder** database

PART D

Check yourself: plasmids

CHECK YOURSELF

1. How many plasmid replicons? On which contigs?
2. Which resistance genes are probably on a **plasmid**? Why does that matter for infection control?
3. Why can’t we be **certain** from an Illumina assembly?

Does the genome match the lab's AST?

- Use the **AST report in your handout** (Part E)
- **It is hypothetical:** written for teaching, not the real AST for this genome
- For each antibiotic: which gene explains it? Do genome and AST **agree**?

PART E

Check yourself: genome vs AST

CHECK YOURSELF

1. Where do genome and AST **disagree**? Give two possible explanations for each.
2. What would you **tell the clinician**? Write one plain-language sentence.
3. Would you use the genome **instead of** AST to treat this patient?

What each tool told us

Tool	Answers
MLST	**Who?** Strain type, known clones
AMRFinderPlus / staramr	**What resistance?** Genes and mutations
PlasmidFinder	**Is it mobile?**
Comparing with AST	**How confident are we?**

Next: what about **many** isolates? → Lecture 4 and Lab 5

Extension for fast pairs

- **ABRicate** with the **card** and **vfdb** (virulence) databases
- Annotate the genome with **Bakta**
- Upload the FASTA to **ResFinder** on the web and compare

Adapted from the GTN tutorial “Identification of AMR genes in an assembled bacterial genome” (CC-BY 4.0).